



The
University of
Faisalabad

Firewall Gateways

Lecture 17



Firewall Gateways

- A **firewall gateway** helps protect your network by checking all the data that comes in or goes out. It uses **proxy programs** to do this job.
- **What Are Proxies?**
- Proxies are like **middlemen** between your computer and the internet.
- They **check and filter** everything going in or out.
- When **data comes in**, it goes to the firewall first.
- When **data goes out**, it looks like it came from the firewall (not your computer).

- **How It Works**
- The firewall has **rules built into the proxy programs.**
- These rules decide what is **allowed** and what is **blocked.**

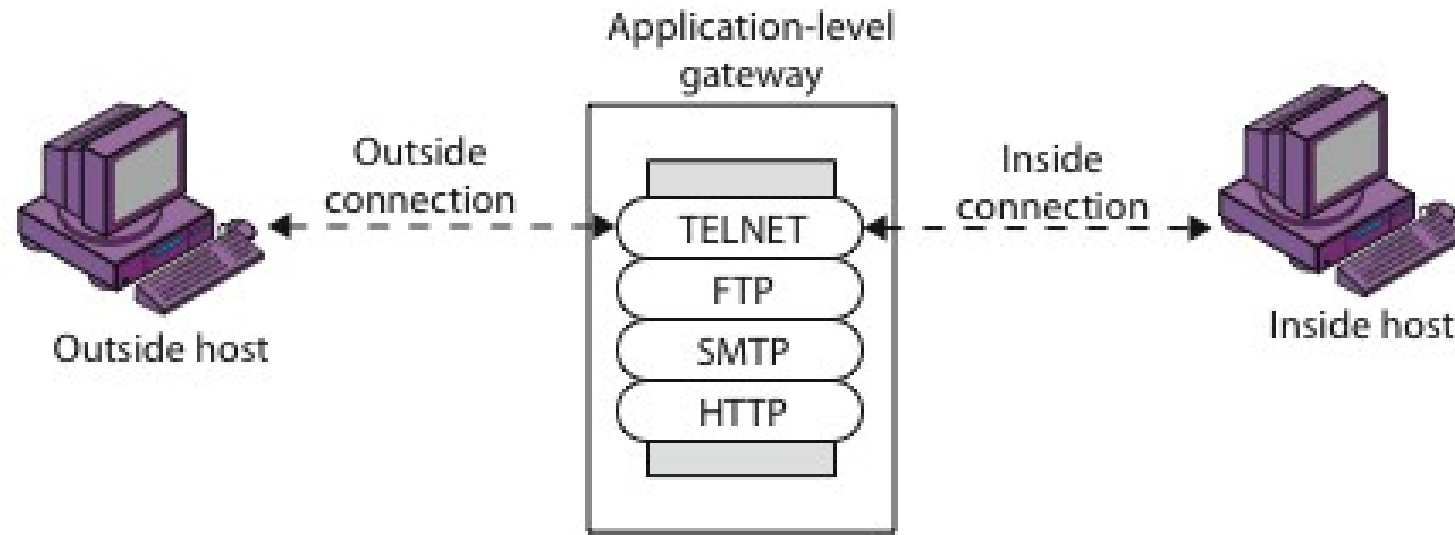
Application-Level Gateways (Proxies)

- These understand **specific types of traffic** like:
 - Web (HTTP)
 - Email (SMTP)
 - File transfers (FTP)
- They are **very smart** and can check inside the content.

Circuit-Level Gateways (Proxies)

- These work at a **lower level**, mainly with **TCP connections**.
- They don't look inside the data, just control the connection.

Firewalls- Application Level Gateway (or Proxy)



(b) Application-level gateway

- An **Application-Level Gateway (ALG)**, also known as a **proxy firewall**, is a type of firewall that works at the **application layer** of the OSI model (Layer 7). It's designed to filter incoming and outgoing traffic **based on specific application-level protocols** like HTTP, FTP, DNS, or SMTP.

What It Does

- An Application-Level Gateway:
- **Acts as an intermediary (proxy)** between the user and the external network.
- **Inspects application-layer data**, not just headers or ports.
- **Filters traffic based on application-specific commands** (e.g., blocking certain HTTP methods like PUT or DELETE).
- Can **perform deep packet inspection (DPI)** to detect malware, viruses, or specific content.

How It Works

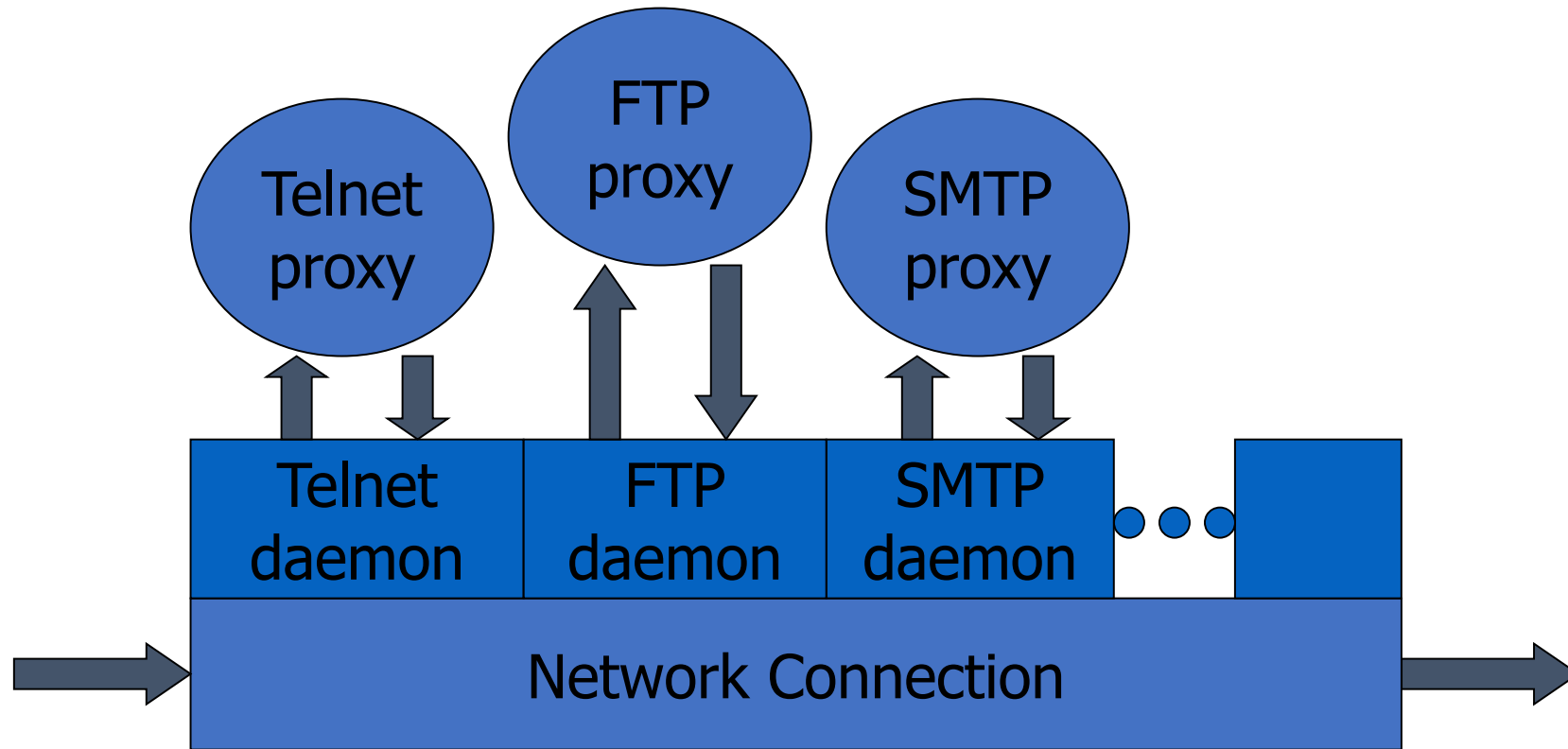
- A client (e.g., your browser) sends a request to connect to a service (like a web server).
- The request is sent to the **ALG**, not directly to the server.
- The ALG **analyzes the request**, makes sure it meets security policies, and then forwards it to the server **on behalf of the client**.
- The server replies to the ALG, which then **forwards the response back to the client**.
- The client and the destination **never communicate directly**—the ALG is always in the middle.

Example Use Case

- Let's say a company uses an ALG for HTTP:
- Employees' web requests go through the proxy.
- The proxy can block access to certain websites (e.g., social media).
- It can scan content for malware or enforce usage policies (e.g., blocking downloads over a certain size).

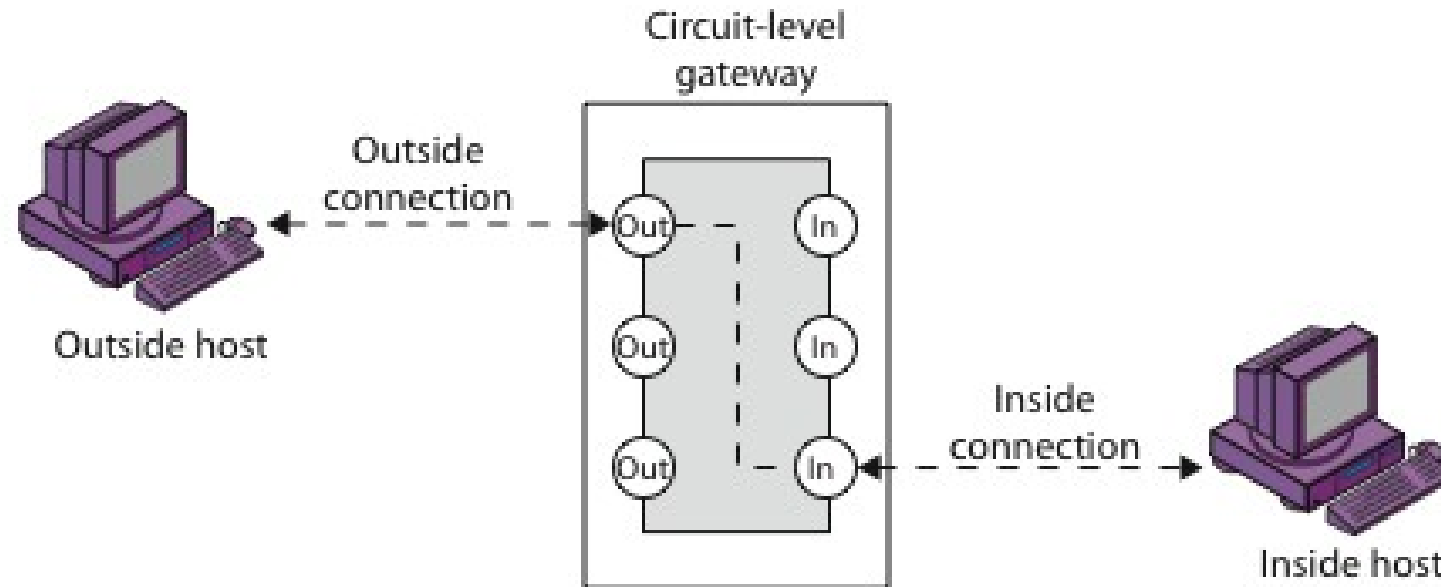
- Has full access to protocol
 - user requests service from proxy
 - proxy validates request as legal
 - then actions request and returns result to user
- Need separate proxies for each service
 - E.g., SMTP (E-Mail)
 - NNTP (Net news)
 - DNS (Domain Name System)
 - NTP (Network Time Protocol)
 - custom services generally not supported

App-level Firewall Architecture



Daemon spawns proxy when communication detected ...

Firewalls- Circuit Level Gateway



(c) Circuit-level gateway

- A **Circuit-Level Gateway** is a type of firewall that **works at the session layer** (Layer 5) of the **OSI model**. It monitors TCP handshakes and session-level communication between internal and external hosts **to ensure that the session is legitimate**—but it doesn't inspect the actual contents of the traffic like an application firewall does.
- Think of it as a **security bouncer** that checks if you have a valid ticket before entering the party—but doesn't care what you do once you're inside.

How It Works

- A client tries to establish a connection to an external server (e.g., a web server).
- The Circuit-Level Gateway **creates a virtual circuit** between the client and the server.
- It **monitors the TCP handshake** to ensure that the connection is legitimate.
- Once the connection is approved, traffic passes through **without being inspected** in detail.
- It typically works using **SOCKS or other proxy protocols**, and **hides internal network details** from the outside world.

Example Use Case

- A corporate firewall uses a Circuit-Level Gateway to allow employees to access external websites.
- The gateway verifies that TCP connections are valid and coming from authorized users.
- Once verified, it forwards packets without inspecting the actual HTTP content.

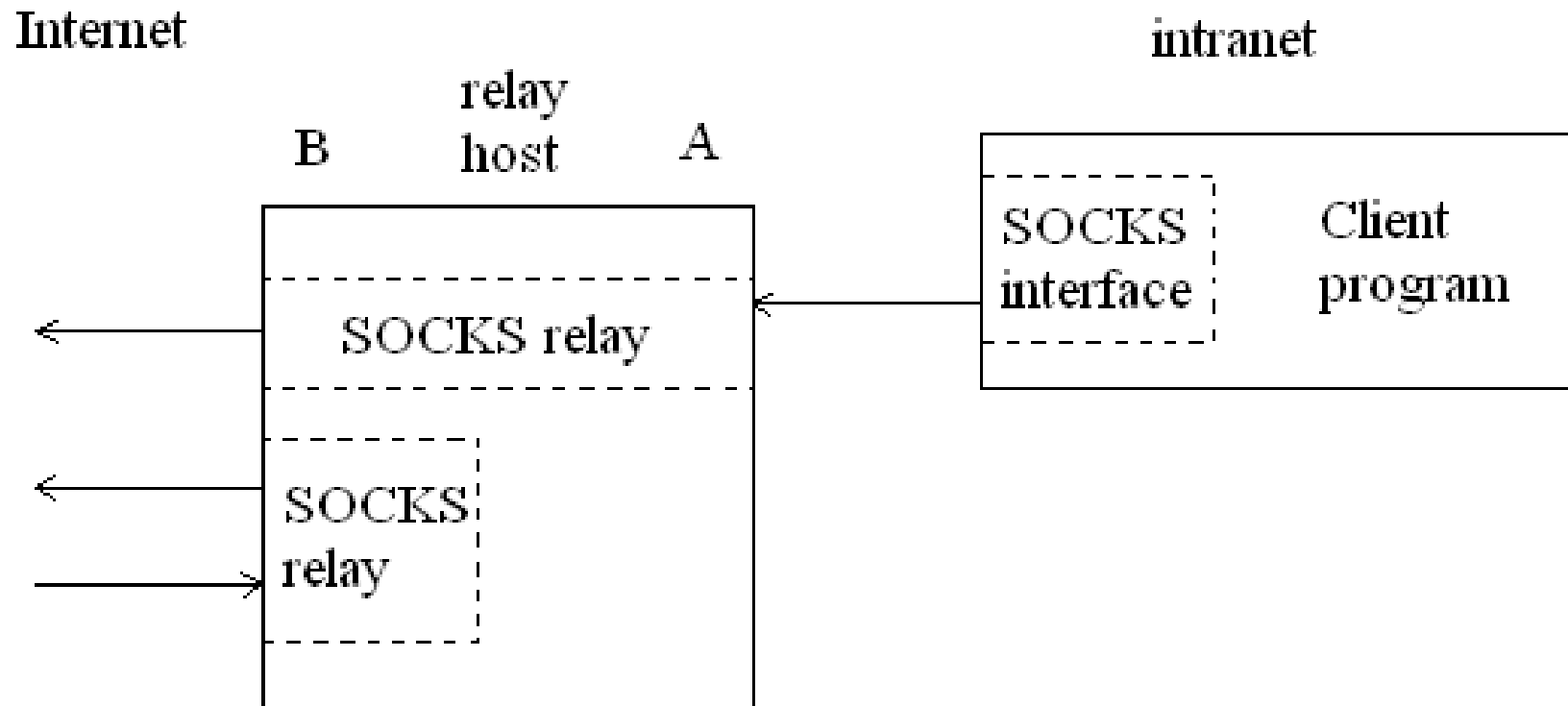


Figure 9.7: A typical SOCKS connection through interface A, and rogue connection through the external interface, B.

Question?